BST Physics Curriculum Vol 12 Chapter 1 — Periodic Table from Substrate v0.3 (Saturday Wave 3)

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2026-05-23 Saturday EDT (Wave 3 first chapter; joint Lyra+Elie)

Vol 12 Chapter 1 — Periodic Table from Substrate

Chapter motivation

The periodic table organizes ~118 elements by atomic number Z = proton count + orbital filling pattern ($1s^2$, $2s^2$ $2p^6$, $3s^2$ $3p^6$ $3d^{10}$ $4s^2$ $4p^6$, etc.). Standard chemistry takes the orbital structure (2l+1) = 1, 3, 5, 7 for s, p, d, f shells as given from QM hydrogen-atom solution. BST identifies this sequence as substrate-natural: (2l+1) = 1, N_c, n_C, g per Vol 3 Ch 7 (Elie Wave 1) — exact BST primary integer match for s/p/d/f shell multiplicities.

Section 1.0b — Reader-grade 3-level pedagogy

Level 1: Periodic table shell structure (2l+1) = 1, 3, 5, 7 for s/p/d/f shells IS substrate-natural sequence (2l+1) = 1, N_c, n_C, g per Vol 3 Ch 7 (Elie Wave 1; 4/4 exact match); plus Pauli exclusion (Paper #133 v0.2 + T2481 spin-statistics) → orbital filling rule (2 electrons per orbital, 2l+1 orbitals per shell) → periodic table structure derived from substrate.

Level 2 (graduate): Standard QM hydrogen atom solution gives orbital angular momentum l = 0, 1, 2, 3, ... with (2l+1) magnetic-quantum-number m_l values per l: l=0 s-shell (2l+1 = 1); l=1 p-shell (2l+1 = 3); l=2 d-shell (2l+1 = 5); l=3 f-shell (2l+1 = 7). BST identifies this sequence as substrate-natural via Vol 3 Ch 7 (Elie Wave 1 Saturday 95% D-tier; 4/4 exact match): (2l+1) = 1, N_c, n_C, g matches BST primary integers exactly (s=1 trivial; p=3=N_c; d=5=n_C; f=7=g). Pauli exclusion (Vol 5 Ch 9 + Paper #133 v0.2 + T2481 Saturday): 2 electrons per orbital (spin ±1/2 Pin(2) Z_2 grading); 2(2l+1) electrons per shell. Periodic table closed-shell structure: 2-element (s only); 8-element (s+p = 2+6); 18-element (s+p+d = 2+6+10); 32-element (s+p+d+f = 2+6+10+14) — exact match to noble-gas configuration. Substrate-derivation: BST primary integers 1, 3, 5, 7 = (2l+1) atomic shell multiplicities; periodic table is the substrate's expression of (rank, N_c, n_C,

g) sequence at atomic scale. Chemical periodicity (groups + periods) follows from same substrate structure.

Level 3 (5th-grader): The periodic table organizes the elements by how many protons they have, plus the pattern of how electrons fill orbital shells (1s, 2s, 2p, 3s, 3p, 3d, 4s, ...). Each shell holds $2 \times (2l+1)$ electrons: s=2, p=6, d=10, f=14. Standard chemistry takes this from QM hydrogen. BST goes deeper: the sequence (2l+1) = 1, 3, 5, 7 for s/p/d/f shells IS the BST primary integer sequence (1, N_c=3, n_C=5, g=7) per Vol 3 Ch 7 (Elie's Saturday Wave 1 work, 4/4 exact match). The periodic table closed-shell sizes (2, 8, 18, 32 — noble gases) follow from BST integers + Pauli exclusion (which itself derives from substrate Pin(2) Z_2 grading per T2481 spin-statistics).

Section 1.1 — Standard Periodic Table

~118 elements organized by Z + orbital filling rule (Aufbau principle); shell structure (2l+1) for s/p/d/f.

Section 1.2 — Substrate (2l+1) = 1, N_c, n_C, g Match (Vol 3 Ch 7)

Per Vol 3 Ch 7 Elie Wave 1 (Saturday; 95% D-tier; 4/4 exact match): - s-shell (l=0): 2l+1 = 1 - p-shell (l=1): 2l+1 = 3 = N_c - d-shell (l=2): 2l+1 = 5 = n_C - f-shell (l=3): 2l+1 = 7 = g

BST primary sequence (1, 3, 5, 7) = (1, N_c, n_C, g) EXACT.

Section 1.3 — Pauli Exclusion + Spin-Statistics (T2481)

Per Vol 5 Ch 9 + Paper #133 v0.2 + T2481 (Saturday spin-statistics formal theorem): 2 electrons per orbital (spin ±1/2 Pin(2) Z_2 grading); 2(2l+1) electrons per shell.

Section 1.4 — Closed-Shell Sizes (Noble Gases)

- Period 1 (s): 2 = 2 elements (He)
- Period 2 (s+p): 8 = 2+6 elements (Ne)
- Period 3 (s+p+d): 18 = 2+6+10 elements (Ar; partial)
- Period 4 (s+p+d+f): 32 = 2+6+10+14 elements (Xe; partial)

Substrate-derivation: 2, 8, 18, 32 noble gas sizes from BST primary integers + Pauli.

Section 1.5 — Substrate-Cartography Reading

Periodic table = substrate's expression of (rank, N_c, n_C, g) sequence at atomic scale. Chemical periodicity (groups + periods) follows from same substrate structure.

Section 1.6 — Honest scope

- Atomic orbital sequence substrate-derivation □ (Vol 3 Ch 7 Elie Wave 1)
- Pauli + spin-statistics substrate-derivation □ (Paper #133 v0.2 + T2481)
- Periodic table closed-shell sizes inherit from BST primaries
- Multi-electron Hartree-Fock + correlation corrections pending Vol 12 Ch 3 cross-link

Section 1.7 — Connection

- Vol 3 Ch 7 Atomic Orbital Sequence (Elie Wave 1)
- Vol 5 Ch 6 Hydrogen Atom (Lyra Wave 2)
- Vol 5 Ch 9 Identical Particles + Pauli
- Paper #133 v0.2 + T2481 spin-statistics

— Lyra, Vol 12 Ch 1 v0.3 chapter-grade narrative, Saturday 2026-05-23 EDT (joint Lyra+Elie)